Band-1 Hadamard Order-2004 Lit-Watch + Paley I Construction

Finite H_2004 construction PASS (prior OPEN errata); full conjecture OPEN; GATE-04 LOCKED

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Document class. Scientific computational / literature ledger increment (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-14_Band1_Hadamard_le64_complete_increment`; `2026-09-11_Band1_Computational_Exper`
Lab. `/workspace/experiments/band1/hadamard_2004_lit_watch/`.
Cutoff. 2026-09-14.

Honesty stamp (non-negotiable).
This memo is a **finite construction / verification increment** for order **2004** only. It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. `clay_closed = false`.

Abstract

Prior Band-1 ledger entries listed Hadamard order **2004** as the smallest commonly cited open admissible order after the August 2026 Alpöge–Voinov–Reynolds-Haertle fill of all gaps ≤ 2000. That status is an **erratum**: Paley type I constructs a Hadamard matrix of order $q+1$ whenever q is a prime power with $q \equiv 3 \pmod 4$. Here $q = 2003$ is prime and $2003 \equiv 3 \pmod 4$, so order **2004** is classical. On-box we construct H_{2004} by Paley I over $\mathbb{F}_{2003}$ and verify $\max |HH^\top - 2004I| = 0$ in exact `int64` arithmetic (**PASS**). Unit tests (H12, H20, Kronecker H24, reject all-ones) all **PASS**. The **full Hadamard conjecture remains OPEN**. Under the standard Sylvester∪Paley∪Kronecker closure of $\{all \leq 2000\}$∪Paley, the next constructive gap is **2036** (dedicated multi-method literature status for 2036 is incomplete this increment).

Keywords. Hadamard matrix; Paley type I; order 2004; literature errata; Band-1; non-claim.

1. Statement lock

Definition. A Hadamard matrix of order n is an $n \times n$ matrix H with entries in $\{\pm 1\}$ satisfying $HH^\top = nI$.

Classical necessary condition. $n \in \{1, 2\}$ or $4 \mid n$.

Finite lemma (this increment only). Explicit on-box Paley type I construction and exact integer certificate for order **2004**. **Not** the Hadamard conjecture.

2. The hunch and the erratum

Hunch (verified). Paley (1933) type I: for prime power $q \equiv 3 \pmod 4$, order $q+1$ is Hadamard. Check $q = 2003$:

Check	Result
2003 prime?	**Yes** (trial division to $\sqrt{2003}$)
2003 mod 4	**3**
Order	$2003 + 1 = 2004$

Prior ledger. `hadamard_le64_complete/literature_ledger.json` recorded `smallest_reported_open_order = 2004`. Secondary sources after the August 2026 ≤ 2000 fill (Cook blog 13 Aug 2026; LavX summary) likewise named 2004 as not yet in S. Those statements overlook the classical Paley I case $q = 2003$.

Likely cause of the popular miss. Cook's illustrative Paley enumeration script starts odd-power exponents at $k = 1$, omitting $p^1 + 1 = p + 1$ for primes $p \equiv 3 \pmod 4$. That silently drops every Paley-I order of the form prime+1, including 2004.

3. Methods

3.1 Paley type I (prime $q \equiv 3 \pmod 4$)

Jacobsthal core $Q_{ij} = \chi(j - i)$ with Legendre symbol χ on $\mathbb{F}_q$. Set $S = Q - I$. Border:

$$H = \begin{pmatrix} 1 & \mathbf{1}^\top \\ \mathbf{1} & S \end{pmatrix}$$

of order $q+1$. (Equivalent skew-Hadamard normalizations exist; we use this bordered form matching the box's prior ≤ 64 verifier.)

3.2 Certificate

Exact NumPy `int64` matmul; require entries in $\{\pm 1\}$ and $\max |HH^\top - nI| = 0$.

3.3 Constructive-gap scan (not a global open-list)

Build the closure of $\{1, 2\} \cup \{4, 8, \ldots, 2000\}$∪Sylvester∪Paley I/II under Kronecker products up to 4000. Report the smallest admissible $n > 2000$ absent from that closure. **This is not** an exhaustive survey of Williamson / Goethals–Seidel / Turyn / other methods.

4. Unit tests

Test	Result
H_{12} Paley I ($q = 11$)	PASS
H_4 Sylvester	PASS
H_{20} Paley I ($q = 19$)	PASS
H_{24} = Kronecker(H_{12}, H_2)	PASS
Reject all-ones 4×4	PASS
$q = 2003$ prime	PASS
$q = 2003 \equiv 3 \pmod 4$	PASS

All 7/7 unit tests PASS.

5. Headline: H_{2004} PASS

Axis	Value
Construction	Paley type I, $q = 2003$
Shape	2004×2004
Entries	$\{\pm 1\}$ only

Axis	Value
$\max \|HH^\top - 2004I\|$	**0**
Construct wall	≈ 2.12 s
Verify wall	≈ 3.96 s
Total wall (script)	≈ 6.08 s
Verdict	**PASS**

Correction. Order **2004** is **constructed and box-certified**. Prior "2004 OPEN" ledger status is **ERRATA**.

6. Literature watch (Aug 2026 context)

Source	What it says about 2004	Our note
Wikipedia *Hadamard matrix* (fetched 2026-09-14)	Aug 2026 Alpöge et al. filled twelve gaps ≤ 2000	Does not assert 2004 open in the fetched text; historical 12-gap list retained
Cook blog 2026-08-13	"not yet known to contain 2004"	**Conflicts** with Paley I at $q = 2003$; treat as secondary erratum
LavX News summary	"conjecture still holds for 2004" / not placed in S	Same overlook of Paley I
Paley 1933 / HandWiki Paley construction	Order $q+1$ for $q \equiv 3 \pmod 4$ prime power	**Applies directly** to $q = 2003$

August 2026 fill (background only). Alpöge–Voinov–Reynolds-Haertle announced explicit ± 1 matrices for the twelve previously open orders ≤ 2000 (668, 716, 892, 1132, 1244, 1388, 1436, 1676, 1772, 1916, 1948, 1964). This increment does **not** re-verify those payloads.

7. Next constructive gap

Under Sylvester $\cup$ Paley $\cup$ Kronecker closure of $\{\text{all } \leq 2000\} \cup$ Paley (limit 4000):

- **2004** $\in$ closure (Paley I).
- First missing admissible: **2036** (then 2052, 2060, 2076, 2092, ...).
- $2036 = 4 \times 509$ has no nontrivial factorization into two Hadamard-admissible factors, and is not Paley I/II for a prime-power q.

Honesty bound. We **do not** claim that 2036 is unresolved by every known method worldwide — only that it is the first gap in this standard constructive closure. Dedicated literature status for 2036 / 2052 / 2060 is **incomplete** this increment.

8. Non-claims

- Full Hadamard conjecture: **OPEN** (Band-3).
- No Clay Millennium problem closed; `clay_closed = false`.
- GATE-04 collection: **LOCKED** (no invented J).
- No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims.
- Finite certificate for **one** order (2004), not for all $4k$.

9. Reproducibility

```
cd /workspace/experiments/band1/hadamard_2004_lit_watch
python3 verify_hadamard_2004.py
```

Emits `results.json`, `literature_ledger.json`, `SUMMARY.txt`.

10. Conclusion

Outcome (A): H_{2004} constructed and verified **PASS**; prior OPEN status for 2004 corrected as **errata**. Full conjecture remains **OPEN**. Next standard-closure gap flagged at **2036** with lit-watch incomplete for non-Paley methods.

Shelf. `2026-09-14_Band1_Hadamard_2004_lit_watch.md` / `.pdf` under `/workspace/DOWNLOAD_PDFs/`.